A novel multi-resolution method using wavelet packet transform and k-means clustering for analyzing data of passive cavitation detectors

Max Au-Yeung and Reza Haqshenas
Ultrasonics Laboratory, Department of Mechanical Engineering, University College London, London, UK

Acoustic emissions from cavitating bubbles contain complex information in the time and frequency domains. Passive cavitation detectors are used to measure these emissions. By processing this data, it becomes possible to extract cavitation features including broadband emissions, harmonics, sub- and ultra-harmonics. This paper presents a novel approach that utilizes the wavelet packet transform, statistical metrics and k-means clustering to detect and classify these features in acoustic emissions. These findings enable us to quantitatively separate the stable and transient features of acoustic cavitation. This method was validated on acoustic emissions obtained by solving a single bubble dynamics (Gilmore-Akulichev model) and subsequently tested on experimental data from biomedical ultrasound applications.

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I. INTRODUCTION

Acoustic cavitation refers to the formation, growth and oscillation of bubbles (cavities filled with gas and vapour) in a liquid using acoustic waves[?] . These bubbles may periodically oscillate whilst exposed to acoustic waves (known as stable cavitation) or grow and implode violently within a few cycles of excitation (known as inertial/transient cavitation). There are several physical effects associated with acoustic cavitation, including creation of free radicals, shock waves, shear stresses, microjets and micro-streaming in the insonated liquid[?] . To utilize these effects for different applications, ultrasound waves are widely used to produce acoustic cavitation, e.g. imaging ultrasound REF, therapeutic ultrasound REF, sono-chemistry, and sono-crystallization REF. To monitor these processes and to perform mechanistic studies, acoustic cavitation must be measured and evaluated quantitatively.

Passive REF and active REF cavitation detections are the two main techniques used to quantitatively measure acoustic emissions generated by bubbles. Passive techniques use a single element ultrasound transducer, i.e. passive cavitation detector (PCD), to acquire acoustic emissions whereas active techniques use more sophisticated ultrasound array transducers. Passive methods are cost-effective and easy to implement, hence they have been widely used in basic and applied research. Some examples in biomedical ultrasound are as follows: thermal ablation REF, histotripsy REF, drug and gene delivery REF, blood-brain-barrier opening REF. Several methods have been developed to analyze acoustic emissions from cavitations, including spectral techniques REFS, combination of spectral and digital filters REFS, time-frequency methods including short time-Fourier transform REF, Bilinear time-frequency distributions (Cohens Class distribution functions) REF, and wavelet transforms REF. These works detect and classify the two main types of acoustic cavitations qualitatively. To carry out fundamental studies and elucidate the concomitant biological/physical effects produced by acoustic cavitation, advanced signal processing techniques are required to detect, extract, classify and quantitatively characterize the features of acoustic cavitation. To tackle this problem, we present a new method which uses combined discrete wavelet packet transform and statistical signal processing for quantitatively analyzing acoustic emissions, extracting and classifying cavitation features. The method is implemented in a Python library which is available in a GitHub repository - see Section 2.2.

The method is discussed in detail in Section II. For testing and verifying the efficacy of the method, synthetic acoustic emissions were generated by solving the problem of single bubble dynamics using Gilmore-Akulichev model, presented in Section 3. The method was then applied to analyze the theoretical acoustic emissions. Section 4 describes the experimental set up, procedure and data processing.

II. MATERIALS AND METHODS

A. Features of acoustic cavitation

The signal from a PCD contains multiple features: (i) a range of harmonics with frequency of nf_0 where f_0 is the fundamental frequency and $n = 1, 2, 3, \ldots$ is an integer. At low powers, the fundamental frequency is the only component[?] . As the ultrasound amplitude is in-

creased, but before collapse cavitation occurs, high-order harmonics are emitted by the bubbles. The emission with frequency equal to twice fundamental was prominent and had an amplitude proportional to the square of the fundamental[?] . (ii) sub-harmonics with frequency of f_0/n. These components can be formed by both stable and inertial cavitation[?] . The main hypothesis about the formation of the sub-harmonics is that they are generated by the non-linear pulsations of stable cavitation. If they are emitted by a system with only inertial cavitation, this can be explained by the prolonged expansion phase that can occur when the bubble oscillates for a few cycles before collapsing. The 1/2 and 1/3 frequency sub-harmonics are the most commonly occurring ones in acoustic cavitation. (iii)ultra-harmonics with frequency of mf_0/n where $m \neq n$ is an integer. These components can be formed by both stable and inertial cavitation[?] . (iv) broadband frequency component which is usually considered as an indicator of transient cavitation. The shock wave produced by an inertial collapse manifests as broadband noise in the spectrum[?] . In the case of a cloud of bubbles, the interaction between bubbles can produce harmonics and sub- and ultra-harmonics as well[?] , further increasing the difficulty in analyzing the PCD data.

These features have different types of time-frequency structures. The (sub- and ultra-) harmonic components of the signal exist for a long time interval but they have localised frequency values equal to mf_0/n for integer m and n values. The transient components, however, have an in impulsive structure in time and relatively broadband structure in the frequency domain. The noise component (mainly measurement noise) exists over a long period and manifests as a broadband distribution in the frequency domain.

B. Literature Review

The problem of classifying cavitation using PCD data has been explored in the past for applications such as ultrasonic cleaning and high-intensity focused ultrasound (HIFU). This section will outline some approaches to the problem of classifying PCD data.

Chitnis et al.[?] note that conventional approaches tend to make use of the Fourier transform alongside a comb-filter. They proposed their own method using decomposition (SVD) to separate transient and harmonic signals. Wu et al. did a review on the acoustic characterisation of cavitation intensity, where the intensity of the cavitation is defined as the energy released by cavitating bubbles per unit time and space. Intensity was identified as a good way to measure cavitation activity. They also note that acoustic methods outperform mechanical, optical, and chemical methods because of its ease of implementation and its relatively low interference with the measured signal. In the context of PCA, the cavitation noise has been shown to have a strong correlation with cavitation intensity.

Liang et al.[?] took PCD data and separated it into its line spectrum and continuous spectrum components to split the data into stable and transient cavitation respectively. This method was based on one that was developed by Frohly et al.[?] who used the integral of the spectrum of the PCD data as a metric to assess its intensity.

Xu et al. identify and assess a similar method, taking the cumulative RMS of a section of the FFT of the signal. This was used to measure the inertial cavitation dose of the inertial cavitation activity. The section of the FFT was chosen between 3.2 and 3.8 MHz to avoid the influence of the harmonics in the signal and was apparently chosen as a heuristic.

Overall, there have been many methods developed to classify cavitation from PCD data but a unified method that can accurately describe the behaviour of the bubbles has yet to be established.

C. Wavelet packet transform

Wavelet packet bases were developed to effectively decompose a piecewise smooth signal with a varying time-frequency structure. Orthonormal wavelet packet bases divide the frequency axis into separate intervals of various sizes, known as scales.

In fast orthogonal discrete wavelet transform (DWT), a bank of conjugate mirror filters and downsampling (by a factor of 2) are used to decompose a signal $x[n]$ of length $N = 2^{n_0}$ into a cascade of coarser approximations $a_j[n]$ and finer details $d_j[n]$ at a scale j [REF]

$$a_{j+1}[n] = \sum_{m=0}^{M-1} h[m]a_j[m-2n] \quad (1)$$

$$d_{j+1}[n] = \sum_{m=0}^{M-1} g[m]a_j[m-2n] \quad (2)$$

where $a_0[n] = x[n]$, $h[m]$ and $g[m]$ are the impulse responses of the filters corresponding to the scaling and wavelet bases functions and are related as follows

$$g[m] = (-1)^{1-m}h[1-m] \quad (3)$$

$g[m]$ and $h[m]$ can be viewed as transfer functions of low- and high-pass filters, respectively, with length M. Thus, approximations and details represent the amplitudes of low- and high- frequency components, respectively, of the decomposed signal. In Eq. 1, the approximations (a_j) are only divided to low- and high-frequency components. Instead, we can iteratively split both approximation (a_j) and detail (d_j) at each scale, usually represented in a binary tree. This increases the computational complexity of discrete wavelet packet transform (DWPT) to $O(Nlog_2(N))$ compared to $O(N)$ for DWT. Any node of the binary tree includes a packet of wavelet transformed component, i.e., the coefficients of filtered signal at a frequency subband, and is labeled by tuple (j, k) where j is the scale number (the depth of the node in the tree) and k is the number of nodes that are on its left at the same scale (position of a node at a given

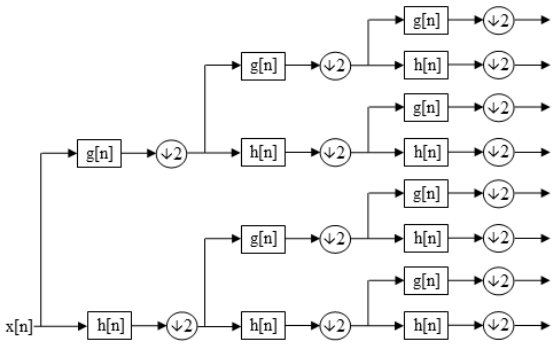


FIG. 1. The DWPT implemented as a filter bank

depth). A schematic diagram of discrete wavelet packet transform using a bank of filters is shown in Fig. 1.

Compared with discrete wavelet transform, DWPT provides higher frequency resolution (and decreased time resolution) at higher frequencies. The more levels of transform, the narrower the frequency subbands. We will use Shannon entropy minimization algorithm REF to find the maximum scales of decomposition for each signal.

The packets can be separately reconstructed into signals which combine to form the original signal. The stable and transient components of the signal tend to have different time-frequency signatures, so the transform will help separate them out. Once separated, the packets can be classified to give an indication of the presence and extent of each type of cavitation.

1. Minimum Entropy Decomposition

Minimum entropy decomposition is an algorithm proposed by Coifman and Wickerhauser[?] . This algorithm uses Shannon entropy defined in equation 6 to decide if a node is worth splitting. If the combined entropy of the child nodes is greater than that of the parent node, the nodes are deemed to have better information and are kept in the final wavelet packet basis. Using this measure of entropy, the algorithm will tend to split strong harmonics further while containing transient features at higher-level nodes. This reduces the redundancy in wavelet packet transform, providing a more compact representation of the signal. It also addresses the class imbalance that comes from the excessive number of nodes at the higher frequencies in the full WPT. This makes use of the primary strength of the WPT in the flexibility it provides in tiling the time-frequency domain.

2. Wavelet Selection

There are many available wavelet families determining the algorithm used to generate the wavelet.

The order of the wavelet determines how many coefficients in the high- and low-pass filters. For example, the number of filter coefficients is equal to twice the wavelet order for Daubechies wavelets. There is a trade-off between the time and frequency resolution when selecting the order of the wavelet. A higher-order wavelet leads to subbands with less overlap but may struggle to pick up closely-spaced features. The effectiveness of wavelets is assessed using the silhouette score in the clustering algorithm.

D. Statistical metrics for feature detection

To extract the signatures of inertial and stable cavitations at each subband, we will use the following statistical metrics: energy, kurtosis, crest factor, entropy and the logarithms of energy. These metrics are evaluated at the final (deepest) scale of DWPT.

The energy of a packet is a measure of how much the signal is concentrated within that subband, calculated as follows

$$E_{j,k} = \sum_{n=1}^{N_{j,k}} x_{j,k}[n]^2 \tag{4}$$

where $x_{j,k}[n]$ are coefficients in a subband (j,k) and $N_{j,k}$ is the number of coefficients in the subband. Note that using orthogonal wavelet bases for DWPT preserves the energy of the signal. Hence, the sum of the energy of all nodes in the final scale equals to the energy of the time series signal.

The average energy of the coefficients in a node is defined as

$$E_k^j = \frac{1}{N_k^j} \mathbf{v}_k^{j\,T} \mathbf{v}_k^j \tag{5}$$

where $\mathbf{v}_k^j$ is a vector containing the coefficients in a subband (j,k) and N_k^j is the number of coefficients in the subband.

To better distinguish fundamental, sub- and ultra-harmonics, we also use the log-energy metric, i.e., $LE_{j,k} = \log E_{j,k}$. To measure the energy distribution of a subband - compared to other subbands, we use Shannon entropy which is defined as

$$S_{j,k} = -\sum_{n=1}^{N_{j,k}} \frac{x_{j,k}[n]^2}{E_{j,k}} \log \frac{x_{j,k}[n]^2}{E_{j,k}} \tag{6}$$

where $E_{0,0}$ is the total energy of the signal.

Level entropy is a variation on the calculation of entropy given by

$$S'_{j,k} = -\sum_{n=1}^{N_{j,k}} \frac{x_{j,k}[n]^2}{E_{0,0}} \log \frac{x_{j,k}[n]^2}{E_{0,0}} \tag{7}$$

Level entropy is used to determine whether a given split provides useful information in the minimum entropy decomposition (see section II C 1).

The spectral entropy is a measure of the complexity of the distribution in the FFT of the subband. This calculated using equation 6 where $x_{j,k}[n]$ is replaced with its magnitude spectrum and $E_{j,k}$ is the energy of the magnitude spectrum. The lower the spectral entropy of a subband, the more deterministic components (information about harmonics) are within the subband. The transient components have a broadband frequency signature, hence a higher entropy can be attributed to transient and/or noise components of the signal. To distinguish these features we need additional statistical metrics.

Kurtosis is a measure of how often outliers occur in a signal. It is viewed as a metric for detecting transient cavitation. Kurtosis is calculated using the following relationship

$$K_{j,k} = \frac{1}{n\sigma_{j,k}^4} \sum_{n=1}^{N_{j,k}} (x_{j,k}[n] - \overline{x}_{j,k})^4 \tag{8}$$

where $\overline{x}_{j,k}$ and $\sigma_{j,k}$ are the mean and standard deviation of the coefficients of the node (j,k).

Skewness measures the extent to which a distribution is asymmetrical. It is given by

$$Sk_{j,k} = \frac{1}{n\sigma_{j,k}^3} \sum_{n=1}^{N_{j,k}} (x_{j,k}[n] - \overline{x}_{j,k})^3 \tag{9}$$

This measure is expected to be larger in magnitude when the signal is more transient. Its sign depends on whether the signal transients tend to be positive or negative. This effect is of no interest so the absolute value of the skewness of the array is used in this study.

The crest factor is the ratio between the peak value and the mean of the signal, indicating how extreme the peaks are in the signal. It is obtained by

$$C_{j,k} = \frac{\max\left(|x_{j,k}[n]|\right)^2}{x_{rms}^2} \tag{10}$$

where x_{rms} is the root-mean-square of $x_{j,k}[n]$.

A feature matrix of these statistical metrics of all subbands is formed. The next step of the algorithm is to classify these features, hence subbands. Different approaches are used for this end. A commonly used technique, adopted in our algorithm too, is k-means clustering[? ?] , which is discussed in the next section.

E. K-means clustering for feature classification

After extracting features, the nodes can classified using methods such as thresholding and clustering. K-means is a simple algorithm used for clustering. It groups a dataset comprised of n observations, each with a number of features, into k clusters. The aim of the algorithm is to minimise the total loss measured as the sum of the Euclidean distances between the centroid of the cluster and each point assigned to it (see equation 11). To do this, the cluster centroids are initiated randomly and each data point is assigned to the nearest cluster. Next, the clusters are re-allocated to minimise the loss function. This process of shifting and reassigning the clusters are repeated until convergence.

$$\Theta = \sum_{i=1}^{n} \sum_{j=1}^{k} \rho^{(i,j)} \|x^{(i)} - \mu^{(j)}\|^2 \tag{11}$$

Where Θ is the distortion (loss) function, ρ is the assignment of the data point, i, to cluster, j, and x and μ are the data point and cluster centroid respectively.

K-means is a non-convex optimisation problem and yields different results depending on the starting position of the clusters. There are several methods that have been developed to combat this, explained by Yiakopulos et al.[?] . Another issue with the k-means algorithm is that the number of clusters must be fixed before training. This presents a challenge in this scenario where the number of clusters is expected to vary between 2-4 depending on the type of cavitation that occurs. Shahapure and Nicholas[?] observed that the silhouette score (see equation 12 is a good metric to determine how many clusters there should be. They employed a simple method which iteratively clustered the data while changing the value of k. The value of k that yields the greatest silhouette score is the selected as the optimal number of clusters. The method employed in this study is further described in section II F 2. Many other techniques for choosing the number of clusters exist **??** but the silhouette coefficient is the simplest to implement and works well for automating the selection of k for smaller datasets.

$$s = \frac{b - a}{\max(a, b)} \tag{12}$$

Where b is the mean distance between a point and the points in the nearest cluster a is the mean distance between the point and all other points in the same cluster. The nearest cluster is found by finding the cluster centre that is closest the the point. The metric used to select k is the average silhouette score for every point in the dataset.

K-means has not been combined with the wavelet packet transform in the context of classsifying cavitation from PCD data, but it has been used in similar in applications such as fault detection and EEG prediction. Both of these domains aim to categorise complex signals containing a mixture transients and harmonics. Yiakopulos et al.[?] applied the k-means algorithm to the problem of fault detection for rolling element bearings. Their method used two instances of K-means, one to determine whether the bearing was faulty and another to determine the nature of the fault. Each algorithm used a different set of used a different of features. They noticed that the non-convexity of k-means was a major disadvantage of the k-means algorithm and identified several techniques

that have been developed to reduce the effect of this. In their method, the centres were initialised using results from a dynamic model of the system. Shete et al.[?] also used the k-means algorithm to detect the k-complex in EEG signals. This was achieved by using the k-means algorithm to compare feature of a reference k-complex signal and extracted sections of an input signal to determine whether that section contained a k-complex. This allows the training of the algorithm by providing it with examples of the feature that is sought after. Desai et al.[?] combined the WPT with k-means to categorise EEG data into different stress states. They calculated statistical features from the reconstructed coefficients at specific subbands and used these as features in the k-means algorithm. This study computes statistical features based on the wavelet coefficients themselves which avoids the computational load introduced by the inverse WPT. This does not cause a drop in performance.

F. Method Overview

The proposed algorithm is outlined in figure 2. This method starts by normalising the data to make sure that the results are consistent. A Blackman-Harris window is applied to reduce the border effects. Next, the DWPT is applied to the signal, decomposing it into frequency subbands using the minimum entropy decomposition algorithm and the and the statistical features are calculated at each node. The nodes are treated as observations with statistical features as attributes and are fed into the k-means algorithm which clusters them.

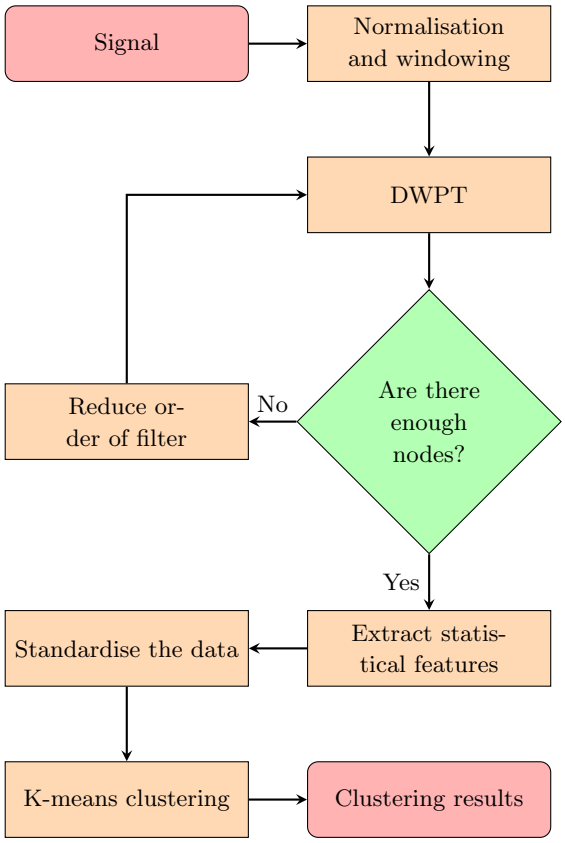


FIG. 2. Flowchart of adaptive method.

1. Adaptive Method

A drawback of this method is its reliance on the minimum entropy decomposition yielding enough nodes for clustering. In this case, a minimum level of decomposition is enforced. This is set to ...

2. Selecting Number of Clusters

Another drawback is that the number of clusters can change depending on the signal. For example, a signal comprised of a sine wave and noise, representing a measurement of stable cavitation, will results in nodes which fall into two clusters: those representing noise, and those representing the fundamental frequency. Once harmonics and transients are added to the signal, more clusters could be formed. Therefore, the algorithm must adapt to the number of clusters that exist in the data. This is done by fitting the clustering algorithm a few times with different numbers of clusters and using the results with the greatest silhouette score.

3. Training Data

The 2nd algorithm involves clustering the nodes from many signals to form a single, more complete, classification model. This model can use more statistical features because of the greater number of observations. If the training data is chosen carefully, it will lead to more meaningful clusters representing harmonics, fundamental frequencies, transients, noise, and combinations of them. As shown in 3, this no longer requires the step of checking if a given signal has enough decompositions and the number of clusters can now be more feasibly chosen manually.

G. Gilmore-Akulichev model

One of them which agrees well with the most of experimental results on the oscillation of a spherical bubble[?] and is valid for high Mach numbers is the Gilmore's model. Gilmore's model takes into account the compressibility of fluid and incorporates sound radiation from an oscillating bubble. The Gilmore equation is expressed as follows[?] :

$$\left(1 - \frac{\dot{R}}{C}\right) R\ddot{R} + \frac{3}{2}\left(1 - \frac{\dot{R}}{3C}\right)\dot{R}^2 = \left(1 + \frac{\dot{R}}{C}\right) H + \frac{\dot{R}}{C}\left(1 - \frac{\dot{R}}{C}\right) R\frac{dH}{dR} \tag{13}$$

where C is the wave speed at the bubble's wall and H is the enthalpy difference in the fluid at pressure P and

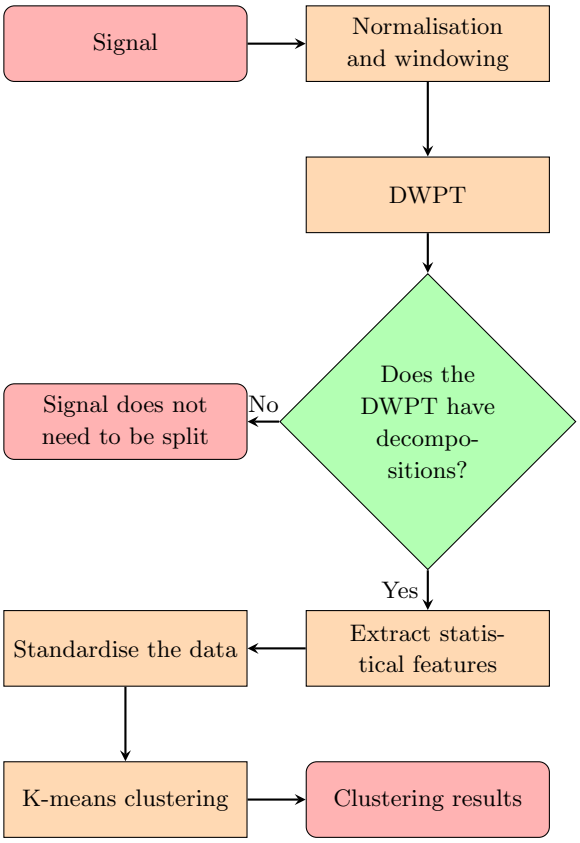


FIG. 3. Flowchart of proposed method.

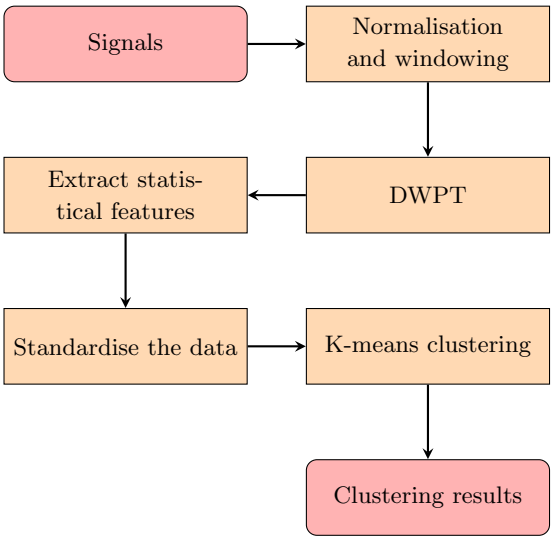


FIG. 4. Non-adaptive method using training data.

P_∞. They are calculated using the following equations:

$$H = \int_{P_\infty}^{P(R)} \frac{dP(\rho_L)}{\rho_L} \tag{14}$$

$$C = \sqrt{c^2 + (n-1)H} \tag{15}$$

where c is the wave speed at far-field. Gilmore used the Tait equation as the equation of state which reads:

$$P_L(\rho) = A\left(\frac{\rho_L}{\rho_{L0}}\right)^n - B \tag{16}$$

substituting Equation 16 into Equation 14, enthalpy can be calculated as a function of the pressure as follows

$$H = \frac{nA^{1/n}}{(n-1)\rho_L}\left[(P(R) + B)^{(1-1/n)} - (P_\infty + B)^{(1-1/n)}\right] \tag{17}$$

where $A, B,$ and n are constants in the Tait equation which are obtained experimentally. In order to determine the enthalpy and the wave speed at the bubble's wall, it is required to calculate the value of pressure in the medium at the surface of the bubble, i.e., $P(R)$. This can be obtained using a modified Laplace equation as follows

$$P(R) = \left(P_0 + \frac{2\sigma}{R_0}\right)\left(\frac{R_0}{R}\right)^{3\kappa} - \frac{2\sigma}{R} - \frac{4\mu\dot{R}}{R} \tag{18}$$

The Gilmore-Akulichev model was solved numerically for different sonication conditions. The implementation and numerical results were verified against published results for identical parameters.

III. RESULTS AND DISCUSSION

This section will present results generated using the proposed method and discuss its performance. These studies aim to test the algorithm's ability to separate the signal into components representing stable and inertial cavitation.

IV. FEATURE SELECTION

The features space is explored using self-organising maps, a type of neural networks which can be used to visualise high dimensional data in 2 dimensions. The algorithm generates nodes which are arranged in a 2 dimensional space, each containing an array of weights, one for each input feature. The nodes are essentially cluster centres which are reliant on each other and determined by a neural network. The features can be plotted separately, giving a map which can be visually compared to evaluate which features might be more useful. The details of the algorithm will not be discussed here and can be found in paper written by Haroun et al.[?] in the context of bearing fault detection and diagnosis.

The data used to train the self-organising map uses 80 signals representing both stable and inertial cavitation. The signals have noise added to them and are windowed and normalised. The minimum entropy decomposition is not used for the first test and a 4-level DWPT is used with a db10 wavelet. The results are shown in 5. When the maps have similar shapes, the features are correlated with each other. This shows that the high energy DWPT nodes tend to have low frequencies, skewness, kurtosis, and spectral entropy, as expected from the physical interpretations of each. This area represents the nodes containing fundamental frequencies. The spectral entropy and log energy are able to find nodes to around the high energy area. These represent harmonics which have less energy than the fundamental frequency but are still picked out using the measure of spectral entropy. Towards the right of the map, there are nodes with high frequency, low kurtosis, and low energy, indicative of noise. At the bottom left, there are nodes with medium frequency, low entropy, and high spectral entropy, indicating nodes containing sharp transients.

Several plots look similar to each other, indicating redundancy in the feature space. Skewness, kurtosis, and crest factor perform similarly, while energy, level entropy (normalised by the energy of the original signal), and log energy are highly correlated because their computations are related.

Next, the minimum entropy decomposition is used with a maximum level of 10 and a minimum level of 3. The level of the node is plotted instead of its centre frequency to highlight the effect of clustering nodes at different levels. The results can be seen in 6. This plot shows that the entropy and spectral entropy are correlated with the level of the node, making them less useful in the minimum entropy decomposition where nodes will be clustered inappropriately depending on their length.

Overall, the three best features are hand picked as average energy, crest factor, and kurtosis as they are independent on the level of the node and have useful physical meanings for classifying nodes that represent stable and transient cavitation.

This method was chosen over PCA which requires the computation of every feature at run time. It is worth noting that the results for self-organising maps is highly inconsistent where the shape of the maps can significantly change depending on the randomly-selected starting points. However, the observed correlations between the features remain similar each time.

A. Analyzing numerical results

In this section, the data is generated using the bubble dynamics model discussed in section II G. The synthetic signals are generated with a sampling rate of 1GHz and are downsampled to a sampling rate of 50MHz. The bubbles are assumed to start at rest, with their radius at a constant size. Next, a ultrasonic signal is applied with frequencies between [0.7 - 1]?MHz and acoustic forcing pressure (P_{AC}) between 10 - 200 kPa. The initial radius

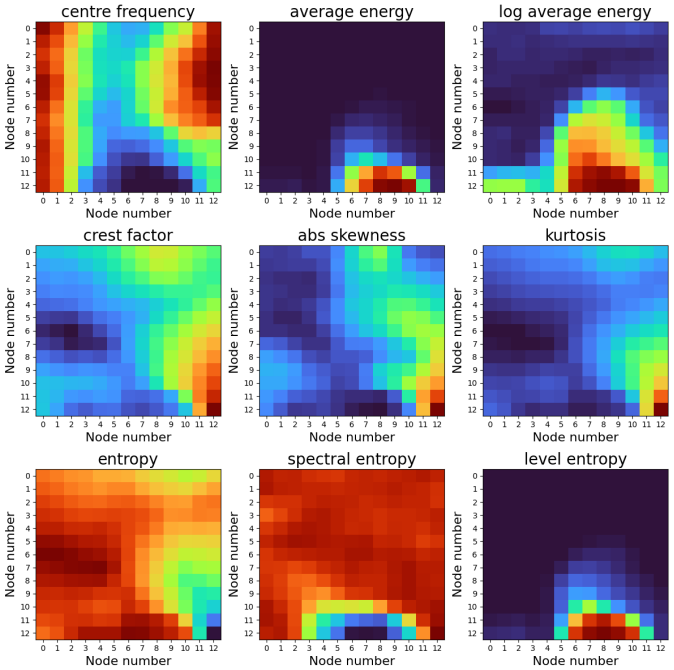


FIG. 5. Self organising maps results where the nodes are arranged in a 12×12 square. Each feature has its own plot to show its relationship with other features. The data was generated using a 4-level DWPT.

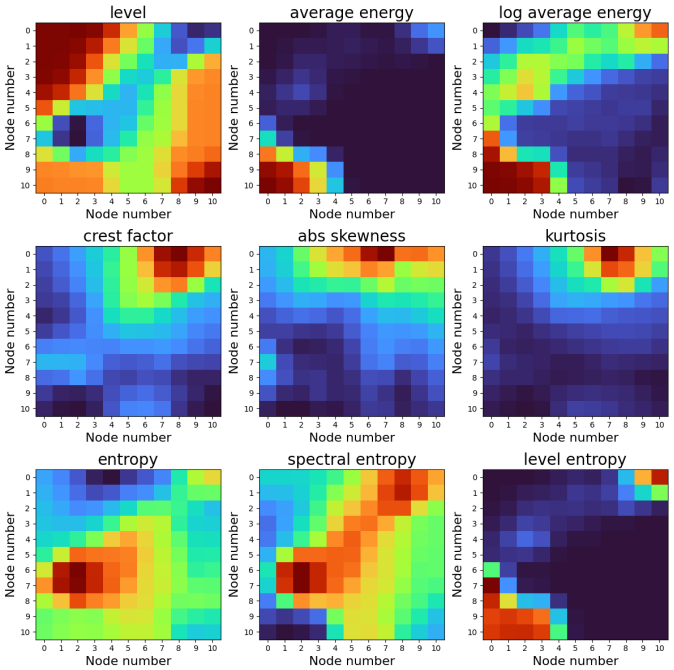


FIG. 6. see caption for figure 5. The data is generated with a minimum entropy decomposition instead of a full 4-level decomposition. The effects of clustering nodes of different lengths can be observed.

of the bubble is also varied between 4 - 11 μm. This ensures a variety of signals to test the data with.

1. Single Signal

The algorithm can be run on a single signal. This approach is useful for characterising a short array of data like that which would be produced in an application such as HIFU. The DWPT is applied to the entire signal, producing a relatively sparse number of nodes for clustering, reducing the accuracy of this method.

Firstly, the The results are shown in figure **??**

B. Experimental set up and protocols

C. Analyzing experimental PCD data

The signal from PCD is expected to contain a range of harmonics[?] . These have frequencies which are multiples of the fundamental frequency (the frequency of the ultrasound). Neppiras outlined the expected acoustic emissions for ultrasound at a range of power levels[?] . At low powers, the fundamental frequency was the only one measured. As the power was increased, but before collapse cavitation occurred, high-order harmonics were emitted by the bubbles. The emission with frequency equal to twice fundamental was prominent and had an amplitude proportional to the square of the fundamental.

Subharmonics occasionally appear. These have frequency of 1/N of the fundamental where N is an integer. This can be formed by both stable and inertial cavitation and there are many theories as to what may cause subharmonics[?] . The main theory is that they are generated by the non-linear pulsations of stable cavitation. If they are emitted by a system with only inertial cavitation, this can be explained by the prolonged expansion phase that can occur when the bubble oscillates for a few cycles before collapsing. The 1/2 and 1/3 frequency sub-harmonics are the most commonly occurring ones in acoustic cavitation.

The final component of the signal is broadband noise. There are several theories as to why this might occur. Most importantly, it is an indicator of transient cavitation. The shock wave is thought to cause this[?] . Another possible cause for broadband noise is the bubble-to-bubble interaction, where bubbles pulsate non-periodically[?] . The interaction between bubbles is thought to produce harmonics and sub-harmonics as well, further increasing the difficulty in analysing clusters of bubbles. Broadband noise has the effect of increasing the intensity of each of the harmonics and sub-harmonics. The transducer will produce some noise as well. The frequency ranges affected depends on the specifications of the transducer. Overall, the feature extraction method must be able to separate and quantify harmonics, subharmonics, and broadband noise. This will make it possible to extract and classify features.

V. SUMMARY

Suggestions for future work:

- finding more appropriate wavelet packet bases.
- exploring alternative algorithms to k-means.
- explore other methods of generating training data with more involved models.
- expanding the work to cavitation in other common mediums like tissue.
- separating noise generated through cavitation and measurement noise.

VI. REFERENCES